RITWIJ ARYAN PARMAR

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Available full-time immediately | MS completed May 2026

AI Systems Engineer specializing in LLM serving runtimes, robotics perception, and high-throughput backend infrastructure.

PROJECTS

HelixServe – LLM Serving Runtime GitHub | Demo

- Built an LLM serving runtime on GCP NVIDIA L4 with paged KV cache, continuous batching, prefix caching, and CUDA Graph decode, increasing mixed-workload throughput from 175.8 to 1,007.8 tokens per second and reducing p95 end-to-end latency from 3.28 seconds to 0.80 seconds for production-style inference.
- Added inference benchmarking and observability for time-to-first-token, inter-token latency, throughput, and KV utilization, achieving 99.8 percent prefix hit rate and reducing repeated-prefix TTFT p95 from 0.299 seconds to 0.158 seconds.

ManoVarta – Controller-Led Multilingual Mental Health GenAI System Live | GitHub | Demo

- Engineered and deployed a public 6-layer multilingual GenAI runtime on Google Cloud Run, integrating web interaction, session orchestration, dialogue planning, LLM-based evidence extraction, structured scoring, and safety routing to convert open-ended English, Hindi, and Hinglish dialogue into PHQ-9 and GAD-7 item assessments.
- Built the bilingual data, evaluation, and benchmarking pipeline end to end, including a 60-session dual-annotated gold dataset and 48 archived conversations replayed through the live runtime endpoint, reaching 78.3 percent coverage completeness, 63.9 percent exact match, 0 parse failures, and 100 percent precision/recall on the evaluated safety set.

SRE-Nidaan – SRE Incident Response Copilot Live | GitHub | Demo

- Built and deployed a 3-microservice SRE incident response copilot on Google Cloud Run using Next.js, FastAPI, and vLLM, integrating 9 APIs for telemetry analysis, runbook retrieval, remediation gating, and analyst feedback to reduce fragmented incident context and support safer operator decisions.
- Aligned an LLM pipeline with QLoRA SFT, reward modeling, and RLHF on 2,500 incident scenarios across 12 SRE domains, adding audit trails and tenant-scoped reliability metrics to make remediation workflows observable, traceable, and reviewable.

TECHNICAL EXPERIENCE

Distributed Robotics and Networked Embedded Sensing Lab, Buffalo, NY Jul 2025 – Present

Research Aide – Robotics Systems Engineering

- Reduced mapping drift to under 1.2 percent across 500 meters of GPS-denied environments by building a ROS2 visual SLAM pipeline for Boston Dynamics Spot with Gaussian Splatting integration.
- Improved LiDAR and VIO fusion for low-texture subterranean navigation, reducing trajectory estimation error by 30 percent while sustaining 20 hertz real-time state estimation.

Tata Elxsi, Bengaluru, India Jan 2024 – Jun 2024

Software Engineer Intern

- Migrated an autonomous vehicle software stack from ROS1 to ROS2 and tuned DDS QoS behavior in CARLA, reducing inter-module latency by 40 percent and achieving sub-100 millisecond communication latency.
- Built an Extended Kalman Filter-based vehicle state estimator that improved positioning accuracy by 35 percent, and implemented a safety-constrained route planner with deterministic state transitions, bounded-latency path generation, and validation hooks, reducing path generation time by 25 percent.

LLMate.ai, Bengaluru, India Apr 2023 – Jul 2023

Backend Engineer Intern

- Built asynchronous backend services with Spring Boot and RabbitMQ for production data workflows, reducing p95 response latency by 40 percent.
- Deployed a GPT-3.5-based text-to-SQL workflow over 50,000 structured records, turning natural-language requests into SQL-backed analytics.
- Set up Docker-based CI/CD with GitHub Actions, improving delivery velocity by 30 percent.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL
AI/ML Systems: LLM Inference, Model Serving, vLLM, Vertex AI, QLoRA, RLHF, CUDA Graphs, Prompt Evaluation
Backend/Infrastructure: FastAPI, Spring Boot, Docker, GCP, Cloud Run, BigQuery, RabbitMQ, Distributed Systems, Observability, CI/CD
Robotics/Autonomy: ROS2, SLAM, LiDAR/VIO Sensor Fusion, CARLA, State Estimation

EDUCATION

University at Buffalo – SUNY Jan 2025 – May 2026
Master of Science in Computer Science and Engineering, GPA: 3.74 out of 4.0

Manipal Institute of Technology Aug 2020 – Aug 2024
Bachelor of Technology in Computer Science and Engineering